Program	B. Tech. (SoCS/SoAE)	Semester	I
Course	Advanced Engineering Mathematics I	Course Code	MATH1059
Session	July-Dec 2025	Unit III	Vector Algebra and Calculus

1. (i) Find unit vector normal to the surface $x^2y + 2xz^2 = 8$ at the point $(1,0,2)$.
 (ii) Find the directional derivative of $\phi = \frac{1}{(x^2+y^2+z^2)^{1/2}}$ at the point $(1,-1,1)$ in the direction of $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}$.
 (iii) Find the directional derivative of $\phi = xy^2 + yz^3$ at $(2,-1,1)$ in the direction of the normal to the surface $x\log_e z - y^2 = -4$ at $(-1,2,1)$.
2. (i) What is the greatest rate of increase of $u = xyz^2$ at the point $(1,0,3)$?
 (ii) In what direction from the point $(3,1,-2)$ is the directional derivative of $\phi = x^2y^2z^4$ maximum and what is its magnitude?
3. Find the values of a, b, c if the directional derivative of $\phi = axy^2 + byz + cz^2x^3$ at $(1,2,-1)$ has maximum magnitude 64 in the direction parallel to the z-axis.
4. Find the direction in which the directional derivative of $\phi = \frac{x^2-y^2}{xy}$ at (1, 1) is zero.
5. Is there a direction u in which the rate of change of the temperature function $T(x,y,z) = 2xy - yz$ at $P(1,-1,1)$ is 3°C/ft? Give reasons for your answer.
6. (i) Find the value of n for which the vector $r^n\vec{r}$ is solenoidal, where $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$.
 (ii) Find the directional derivative of the divergence of $\vec{F} = xy\hat{\imath} + xy^2\hat{\jmath} + z^2\hat{k}$ at the point $(2,1,2)$ in the direction of the outer normal to the sphere $x^2 + y^2 + z^2 = 9$.
7. (i) Find $curl(curl\,\vec{A})$, if $\vec{A} = x^2y\,\hat{\imath} - 2xz\,\hat{\jmath} + 2yz\,\hat{k}$ at the point $(1,0,2)$.
 (ii) Determine the constants a and b if $\vec{F} = (2xy + 3yz)\hat{\imath} + (x^2 + axz - 4z^2)\hat{\jmath} + (3xy + 2byz)\hat{k}$ is irrotational.
 (iii) Prove that $\vec{A} = (6xy + z^3)\,\hat{\imath} + (3x^2 - z)\,\hat{\jmath} + (3xz^2 - y)\,\hat{k}$ is irrotational. Find the function ϕ such that $\vec{A} = \nabla\phi$.
8. Prove that for every field $A, div\,(curl\,A) = 0$.
9. Can a fluid flow described by the vector field $\vec{F} = x^2y\,\hat{\imath} + y - xy^2\hat{\jmath}$ be a magnetic field? (Hint: For a magnetic field's divergence must be zero.)
10. Evaluate $\int_C \vec{F}.d\vec{r}$, where $\vec{F} = \cos y\,\hat{\imath} + x\sin y\,\hat{\jmath}$ and C is curve $y = \sqrt{1-x^2}$ in the xy-plane from $(1,0)$ to $(0,1)$.
11. Evaluate $\int_C \vec{F}.d\vec{r}$, where $\vec{F} = (x+y)\,\hat{\imath} + (y-x)\,\hat{\jmath}$ and C is
 (i) the parabola $y^2 = x$ between the points $(1,1)$ and $(4,2)$
 (ii) the straight line joining the points $(1,1)$ and $(4,2)$.

12. Find the work done in moving a particle along the straight line segments joining the points (0,0,0) to (1,0,0), then to (1,0,0) and finally to (1,1,1) under the force field $\vec{F} = (3x^2 + 6y)\hat{\imath} - 14yz\hat{\jmath} + 20xz^2\hat{k}$.

13. Find the work done by the force $\vec{F} = x\hat{\imath} - z\hat{\jmath} + 2y\hat{k}$ in displacing the particle along the triangle OAB, where

$$\text{OA}: 0 \leq x \leq 1\,, y = x, z = 0$$

$$\text{AB}: 0 \leq z \leq 1, x = 1, y = 1$$

$$\text{BO}: 0 \leq x \leq 1\,, y = z = x.$$

14. Find the work done in moving a particle in a force field $\vec{F} = (2x - y + z)\hat{\imath} + (x + y - z^2)\hat{\jmath} + (3x - 2y + 4z)\hat{k}$ once around the circle in xy-plane with centre at the origin and radius of 3 units.

15. Prove that the line integral of $F = (y\,sin\,z - sin\,x)\,\hat{\imath} + (x\,sin\,z + 2yz)\hat{\jmath} + (xy\,cos\,z + y2\,)\hat{k}$ is independent of the path of integration and hence, find its scalar potential ϕ.

16. If $\vec{F} = (2xy + z^3)\hat{\imath} + x^2\hat{\jmath} + 3xz^2\hat{k}$ is conservative, then

a. find its scalar potential ϕ

b. find the work done in moving a particle under this force field from $(1, -2, 1)$ to $(3, 1, 4)$.

17. Find the work done in moving a particle in a force field $\vec{F} = 3x^2\hat{\imath} + (2xz - y)\hat{\jmath} + z\hat{k}$ along the

(i) straight line joining the points $(0,0,0)$ and $(2,1,3)$

(ii) curve $x = 2t^2, y = t, z = 4t^2 - t$ from $t = 0$ to $t = 1$.

18. Verify Green's theorem for $\oint_C [(x^2 - 2xy)dx + (x^2y + 3)dy]$ where C is the boundary of the region bounded by the parabola $y = x^2$ and the line $y = x$.

19. Evaluate $\oint_C [(x^2 + 2y)dx + (4x + y^2)dy]$ by Green's theorem where C is the boundary of the region $y = 0, y = 2x$ and $x + y = 3$.

20. Find the area of the region bounded by the parabola $y = x^2$ and the line $y = x + 2$ using Green's theorem.

21. Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ using Green's Theorem.

22. Use the Green's Theorem to find the area enclosed by the circle $\vec{r}(t) = (a\,cos\,t)\hat{\imath} + (a\,sin\,t)\hat{\jmath},\ 0 \leq t \leq 2\pi$.

23. Evaluate $\iint_S (yz\,dy\,dz + xz\,dz\,dx + xy\,dx\,dy)$ over the surface of the sphere $x^2 + y^2 + z^2 = 1$ in the positive octant.